

Ligjerata 7ë

Masa e Diellit dhe Tokës

I)

$$\frac{T^2}{r^3} = \frac{4\pi^2}{\gamma M_D}$$

$$r = 150 \cdot 10^6 \text{ km}$$

$$T = 365 \text{ ditë}$$

$$\gamma = 6.63 \cdot 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$M_D = \frac{4\pi^2}{\gamma T^2} \cdot r^3$$

$$M_D = \left(\frac{2\pi}{T}\right)^2 \cdot \frac{r^3}{\gamma}$$

$$M_D = 1.99 \cdot 10^{30} \text{ kg}$$

II)

$$g_\oplus = \gamma \frac{M_T}{R_T^2}$$

$$g_\oplus \cdot R_T^2 = \gamma M_T / : \gamma$$

$$M_T = \frac{g_\oplus R_T^2}{\gamma} = 5.96 \cdot 10^{24} \text{ kg}$$

$$R_T = 6.38 \cdot 10^6 \text{ m}$$

$$g_\oplus = 9.81 \frac{\text{m}}{\text{s}^2}$$

$$\gamma = 6.63 \cdot 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$